

J. Garrett Morris

DEPARTMENT OF COMPUTER SCIENCE
14 MacLean Hall // 2 West Washington Street // Iowa City, Iowa 52242
garrett-morris@uiowa.edu // <https://jgbm.github.io>

Academic Appointments

THE UNIVERSITY OF IOWA
Associate Professor
Assistant Professor

IOWA CITY, IOWA
July 2026–present
August 2020–June 2026

THE UNIVERSITY OF KANSAS
Assistant Professor

LAWRENCE, KANSAS
August 2017–July 2020

Education and Professional Preparation

THE UNIVERSITY OF EDINBURGH
Post-doctoral Research Associate

EDINBURGH, SCOTLAND
October 2013–July 2017

PORTLAND STATE UNIVERSITY
Post-doctoral Research Associate

PORTLAND, OREGON
June 2013–September 2013

PORTLAND STATE UNIVERSITY
Ph.D. in Computer Science.
Dissertation: *Type Classes and Instance Chains: A Relational Approach*
Adviser: Mark P. Jones

PORTLAND, OREGON
June 2013

DARTMOUTH COLLEGE
A.B. with High Honors in Computer Science.
Thesis: *A Binding Time Analysis Technique for Functional Programs*
Adviser: Doug McIlroy

HANOVER, NEW HAMPSHIRE
June 2005

Publications

J. Garrett Morris. “Another Type Inference Algorithm for First-Class Implicit Polymorphism”. In *Proc. ACM Prog. Lang.* 10 (ICFP) (September 2026).

Wenhao Tang, Daniel Hillerström, Sam Lindley, and J. Garrett Morris. “Soundly Handling Linearity”. In *Proc. ACM Prog. Lang.* 8 (POPL) (January 2024). Distinguished paper.

Simon Fowler, Wen Kokke, Ornela Dardha, Sam Lindley, and J. Garrett Morris. “Separating Session Smoothly”. In *Log. Methods Comput. Sci.* 19 (3) (2023).

Alex Hubers and J. Garrett Morris. “Generic Programming with Extensible Data Types: Or, Making Ad Hoc Extensible Data Types Less Ad Hoc”. In *Proc. ACM Prog. Lang.* 7 (ICFP) (September 2023).

Pedro Abreu, Benjamin Delaware, Alex Hubers, Christa Jenkins, J. Garrett Morris, and Aaron Stump. “A Type-Based Approach to Divide-and-Conquer Recursion in Coq”. In *Proc. ACM Prog. Lang.* 7 (POPL) (January 2023).

Apoorv Ingle, Alex Hubers, and J. Garrett Morris. “Partial type constructors in practice.”. In *Proceedings of the 15th ACM SIGPLAN International Symposium on Haskell*, Haskell ’22, Ljubljana, Slovenia, 2022.

Simon Fowler, Wen Kokke, Ornela Dardha, Sam Lindley, and J. Garrett Morris. “Separating Sessions Smoothly”. In *Proceedings of the 32nd International Conference on Concurrency Theory*, volume 203 of *Leibniz International Proceedings in Informatics*, Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2022.

Mark P. Jones, J. Garrett Morris, Richard Eisenberg. “Partial Type Constructors; or, making ad hoc data types less ad hoc”. In *Proc. ACM Prog. Lang.* 4, POPL (January 2020).

Wen Kokke, J. Garrett Morris, Philip Wadler. “Towards Races in Linear Logic.” In *Coordination Models and Languages - 21st IFIP WG 6.1 International Conference*, volume 11533 of *Lecture Notes in Computer Science*, pages 37–53, Springer, 2019.

J. Garrett Morris, James McKinna. Abstracting Extensible Data Types; or, rows by any other name. In *Proc. ACM Prog. Lang.* 3, POPL (January 2019).

Simon Fowler, Sam Lindley, J. Garrett Morris, Sára Decova. Exceptional asynchronous session types: session types without tiers. In *Proc. ACM Prog. Lang.* 3, POPL (January 2019).

Jack Williams, J. Garrett Morris and Philip Wadler. The Root Cause of Blame: Contracts for Intersection and Union Types. In *Proc. ACM Prog. Lang.* 2, OOPSLA (October 2018).

J. Garrett Morris and Richard A. Eisenberg. Constrained Type Families. In *Proc. ACM Program. Lang.* 1, ICFP, Article 42 (August 2017), 28 pages.

Jack Williams, J. Garrett Morris, Philip Wadler, and Jakub Zalewski. Mixed Messages: Measuring Conformance and Non-Interference in Type Script. In *Proceedings of the 31st European Conference on Object-Oriented Programming, ECOOP '17*, Barcelona, Spain, 2017. Schloss Dagstuhl–Leibniz-Zentrum für Informatik.

Sam Lindley and J. Garrett Morris. Lightweight Functional Session Types. In Simon Gay and Antonio Ravara (Eds.), *Behavioral Types: from Theory to Tools.*, River Publishers, Denmark, 2017.

J. Garrett Morris. The Best of Both Worlds: Linear Functional Programming without Compromise. In *Proceedings of 21st ACM SIGPLAN international conference on Functional Programming, ICFP '16*, Nara, Japan, 2016. ACM.

Sam Lindley and J. Garrett Morris. Talking Bananas: Structural Recursion for Session Types. In *Proceedings of 21st ACM SIGPLAN international conference on Functional Programming, ICFP '16*, Nara, Japan, 2016. ACM.

Sam Lindley and J. Garrett Morris. Embedding Session Types in Haskell. In *Proceedings of the 9th ACM SIGPLAN symposium on Haskell, Haskell '16*, Nara, Japan, 2016. ACM.

Robert Atkey, Sam Lindley, and J. Garrett Morris. Conflation Confers Concurrency. In *A List of Successes that can Change the World - Essays Dedicated to Philip Wadler on the Occasion of his 60th birthday*, volume 9600 of *Lecture Notes in Computer Science*, pages 32–55, Springer, 2016.

J. Garrett Morris. Variations on Variants. In *Proceedings of the 8th ACM SIGPLAN symposium on Haskell, Haskell '15*, Vancouver, British Columbia, 2015. ACM.

Sam Lindley and J. Garrett Morris. A Semantics for Propositions as Sessions. In *24th European Symposium on Programming, ESOP '15*, London, UK, 2015. Springer.

J. Garrett Morris. A Simple Semantics for Haskell Overloading. In *Proceedings of the 7th ACM SIGPLAN symposium on Haskell, Haskell '14*, Gothenburg, Sweden, 2014. ACM.

Sam Lindley and J. Garrett Morris. Sessions as Propositions. In *PLACES 14*.

J. Garrett Morris and Mark P. Jones. Instance chains: Type-class programming without overlapping instances. In *Proceedings of the 15th ACM SIGPLAN international conference on Functional programming, ICFP '10*, Baltimore, Maryland, 2010. ACM.

J. Garrett Morris. Experience report: Using Hackage to inform language design. In *Proceedings of the third ACM symposium on Haskell, Haskell '10*, Baltimore, Maryland, 2010. ACM.

Invited Presentations

- “Partial Type Constructors”, at the Midwest Programming Languages Summit, West Lafayette, Indiana, 2019.
- “On Extensibility”, at the University of Iowa Computer Science Colloquium, Iowa City, Iowa, 2018.
- “On Unsatisfiability”, at the Midwest Programming Languages Summit, Madison, Wisconsin, 2018.
- “On Unsatisfiability”, at the Haskell Implementers Workshop, Oxford, UK, 2017.
- “A Row By Any Other Name”, at a meeting of IFIP Working Group 2.8, Edinburgh, UK, 2017.
- “Semantical Analysis of Type Classes”, at the Workshop on Coalgebra, Horn Clause Logic Programming, and Types, Edinburgh, UK, 2016.
- “The Best of Both Worlds: Linear Functional Programming without Compromise”, at a meeting of COST action CA15125 (EUTYPES), Lisbon, Portugal, 2016.
- “Embedding Session Types in Haskell”, at a meeting of COST action IC1201 (Behavioral Types), Lisbon, Portugal, 2016.
- “Substructural Types with Class”, at the Workshop on Type Inference and Automated Proving, Dundee, UK, 2015.
- “Lightweight Functional Session Types in Links”, at the Workshop on Communication and Concurrency: Practicalities of Programming with Sessions, Glasgow, UK, 2015.
- “Propositions as Sessions, Semantically”, University of Glasgow FATA (Formal analysis, theory and algorithms) seminar, Glasgow, UK, 2015.
- “Recursive Session Types and Linear Logic”, at a meeting of COST action IC1201 (Behavioral Types), Grenoble, France, 2014.

External Funding

- “Travel: Midwest Programming Language Summit 2026–2028,” Principal Investigator. National Science Foundation. Award total: \$15,000.
- “SHF: MEDIUM: Extensible LCF-style Theorem Proving with Dependent Rows,” Principal Investigator. National Science Foundation. 2025–2029. Award total: \$1,199,815.
- “Collaborative Research: FMitF: Track III: Integrating Formal Methods into the Foundational Undergraduate Curriculum,” Principal Investigator. National Science Foundation. 2024–2027. Award total: \$77,430.
- “SHF: SMALL: A New Semantics for Type-Level Programming in Haskell”, Principal Investigator. National Science Foundation. 2024–2027. Award total: \$599,291.
- “CAREER: Extensibility in Theory and Practice”, Principal Investigator. National Science Foundation. 2021–2027. Award total: \$524,614.
- “Science of security for cyber-physical systems”, Co-principal Investigator, with Perry Alexander (PI), Alex Bardas, Fengjun Li, Bo Luo, Prasad Kulkarni, John Symons, Heechul Yun. National Security Agency. 2017–2020. Award total: \$14,752,121.

National and International Service

Program committee, 2027 ACM SIGPLAN International Conference on Functional Programming, Nijmegen, the Netherlands, 2027.

Organizer, Midwest Programming Languages Symposium, 2026.

Steering committee, Haskell Symposium, 2024–2028.

Co-chair, Haskell Symposium, 2025.

Program committee, 2025 ACM SIGPLAN Symposium on the Principles of Programming Languages, Denver, CO, 2025.

Chair, Haskell Symposium, 2024.

Reviewer, ACM Transactions on Programming Languages and Systems, 2018, 2020–2021.

Reviewer, Journal of Functional Programming, 2019, 2021.

Program committee, 2022 ACM SIGPLAN Symposium on the Principles of Programming Languages, Philadelphia, PA, 2022.

Program committee, 2021 ACM SIGPLAN Workshop on Partial Evaluation and Program Manipulation, Copenhagen, Denmark, 2021.

External reviewer, 2020 ACM SIGPLAN International Conference on Functional Programming, Jersey City, NJ, 2020.

External reviewer, 32nd International Conference on Concurrency Theory, Paris, France, 2020.

External reviewer, 2020 European Symposium on Programming, Dublin, Ireland, 2020.

Program committee, 2019 Workshop on Type-driven Development, Berlin, Germany, 2019.

Program committee, 2019 European Symposium on Programming, Prague, Czech Republic, 2019.

Program committee, 30th Symposium on Implementation and Application of Functional Languages, Boston, MA, 2018.

Program committee and distinguished paper committee, 2018 ACM SIGPLAN International Conference on Functional Programming, St. Louis, MO, 2018

Program committee, 2017 Workshop on New Object-Oriented Languages, Vancouver, BC, 2017.

Program committee, 29th Symposium on Implementation and Application of Functional Languages, Bristol, UK, 2017.

Program committee, 2017 ACM SIGPLAN Symposium on Haskell. Oxford, UK, 2017.

Program committee, Workshop on Coalgebra, Horn-Clause Logic Programming, and Types, Edinburgh, UK, 2016.

Program committee, 2016 Workshop on New Object-Oriented Languages, Amsterdam, 2016.

Program committee, 2015 ACM SIGPLAN Symposium on Haskell. Vancouver, BC, 2015.

Organizer, Communication and Concurrency: Practicalities of Programming with Sessions (with Roly Perera, Dimitrios Kouzapas). Glasgow, 2015.

University and Departmental Service

At the University of Iowa

Executive committee, Department of Computer Science, 2024–2026.

Departmental representative, College of Liberal Arts and Sciences Faculty Assembly, 2024–2025.

Undergraduate committee, Department of Computer Science, 2021–2024.

Honors program coordinator, Department of Computer Science, 2021–2023.

Department secretary, Department of Computer Science, 2020–2021.

At the University of Kansas

Academic misconduct panel (chair), School of Engineering, 2019.

Committee on faculty rights and responsibilities, Department of Electrical Engineering and Computer Science, 2017–2020.

Student awards committee, Department of Electrical Engineering and Computer Science, 2017–2020.

Supervision

Postdoctoral Researchers

Andrew Marmaduke, University of Iowa, 2024–present.

Doctoral students

Susan Harding, University of Iowa, 2026–present.

Oscar Bender-Stone, University of Iowa, 2026–present.

Tristan MacKinlay, University of Iowa, 2025–present.

Jason Veenendaal, University of Iowa, 2024–present.

Alex Hubers, University of Iowa, 2022–present.

Apoorv Ingle, University of Iowa and University of Kansas, 2018–present.

Andrei Elliot, University of Kansas, 2019–2020.

Wen Kokke, 2016–2017 (Co-supervisor, with P. Wadler, lead supervisor).

Jack Williams, 2015–2017 (Co-supervisor, with P. Wadler, lead supervisor).

Jakub Zalewski, 2016–2017 (Co-supervisor, with P. Wadler, lead supervisor).

Master's students

Dustin Wendt, University of Kansas, 2018–2020. Topic: “Formalizing Session Types and their Semantics”.

Apoorv Ingle, University of Kansas, 2017–2018. Topic: “Resource Aware Functional Languages”.

Jack Williams, MSc. Computer Science, University of Edinburgh, 2014. (Sole supervisor, before he undertook Ph.D. studies.) Topic: “Concurrent Data Structures in Linear Logic”.

Undergraduate students

Emily P. Johnson, University of Iowa, 2020–2021.

John Quitno, University of Kansas, 2018–2019.

Emilia Paz Ojeda, University of Kansas, 2018–2019.

Luke Dercher, University of Kansas, 2018.

Brian Quiroz, University of Kansas, 2019.

Alex Shadley, University of Kansas, 2019–2020.

Trenton Potter, University of Kansas, 2019–2020.

Instruction**At the University of Iowa**

COURSE	SEMESTER	ENROLLMENT
CS:3820—Programming Language Concepts	Fall 2026	55
CS:3820—Programming Language Concepts	Spring 2026	110
CS:4980—Topics II (Functional algorithms)	Spring 2025	14
CS:5850—Programming Language Foundations	Spring 2024	9
CS:3820—Programming Language Concepts	Fall 2023	62
CS:4980—Topics II (Functional algorithms)	Spring 2023	15
CS:3820—Programming Language Concepts	Fall 2022	98
CS:5850—Programming Language Foundations	Spring 2022	10
CS:3820—Programming Language Concepts	Fall 2021	110
CS:4980—Topics II (Functional algorithms)	Spring 2021	19
CS:3820—Programming Language Concepts	Fall 2020	47

At the University of Kansas

COURSE	SEMESTER	ENROLLMENT
EECS 662—Programming Languages	Spring 2020	41
EECS 210—Discrete Structures	Spring 2020	24
EECS 762—Programming Language Foundations I	Fall 2019	4
EECS 662—Programming Languages	Spring 2019	31
EECS 762—Programming Language Foundations I	Fall 2018	19
EECS 665—Compiler Construction	Spring 2018	14
EECS 662—Programming Languages	Fall 2017	32

Other Employment

ADVANCED MICRO DEVICES
Co-op engineer

SUNNYVALE, CALIFORNIA
June 2010–September 2010

RAYBEAM SOLUTIONS, INC.
Independent contractor

BOSTON, MASSACHUSETTS
December 2007–August 2008

AUTONOMY
Consultant

SAN FRANCISCO, CALIFORNIA
February 2007–November 2007

AETION TECHNOLOGIES
Software Architect

COLUMBUS, OHIO
August 2005–November 2006

DARTMOUTH COLLEGE
Undergraduate Teaching Assistant

HANOVER, NEW HAMPSHIRE
January 2002–June 2005

Last updated: September 1, 2026
<http://jgbm.github.io/cv.pdf>